

LendingClub Interest Rate Prediction

Gradient Boosting + Neural Network Ensemble

Team [Number] | [Member Names]
Section: [Your Section] | Group: [Your Group Number]

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Problem Statement & Approach

Business Problem

- Predict loan interest rates using only origination-time info

Our Approach

- Five models: 3 Gradient Boosting + 2 Neural Networks
- Temporal validation to simulate real-world deployment
- SHAP + Permutation Importance for explainability

Target: int_rate (mean=12.95%, range=6.46%-30.99%)

100K training, 10K test, 36 origination features

KEY: Prevent data leakage. Removed: loan_status (post-origination outcome).

Data Processing & Anti-Leakage

Features Removed (Leakage Risk)

- loan_status: Post-origination outcome
- title: Redundant with purpose field

Feature Engineering (29 new from 36 original)

- FICO midpoint + risk categories, credit utilization, loan-to-income
- Missingness indicators, delinquency/bankruptcy flags
- Employment title grouping (35K unique -> 40 + Other)

Temporal Split: 70% train (oldest) / 15% val / 15% test (newest)

All preprocessing fit on training data only

Modeling: 5 Models + Ensemble

Gradient Boosting

- XGBoost: Histogram-based, regularized
- LightGBM: Leaf-wise tree growth
- CatBoost: Ordered boosting, native categoricals

Neural Networks

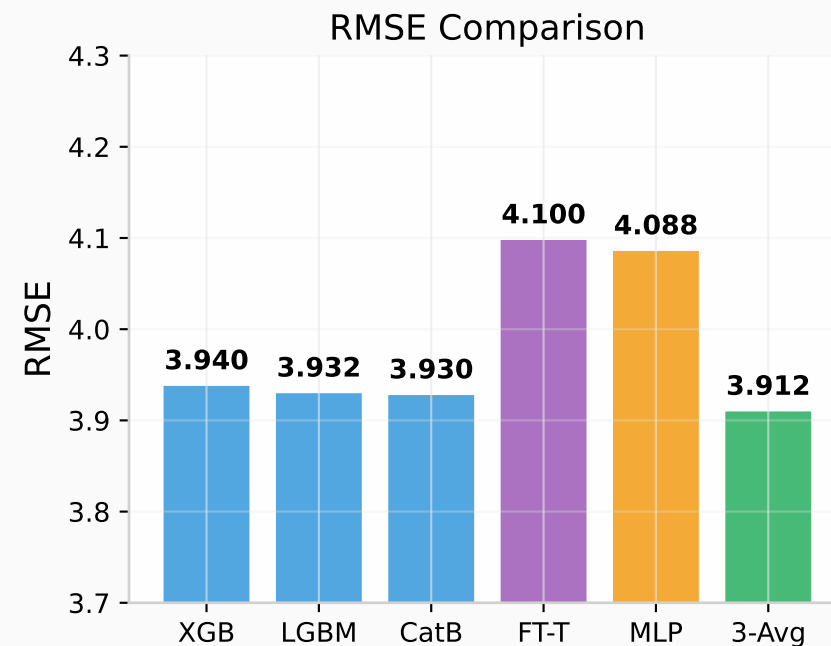
- FT-Transformer: Feature Tokenizer +
- Transformer Encoder (Gorishniy 2021)
- Tab-MLP: Deep MLP with LayerNorm + GELU

Ensemble: 3-Model GBM Average (Best)

- Simple average of XGBoost, LightGBM, CatBoost
- Outperforms all individual models and neural networks
- Validation: Temporal split (no random shuffle)

Model Comparison: GBM vs. Neural Networks

Model	RMSE	MAE	MAPE	R2
XGBoost	3.940	2.961	23.76%	0.448
LightGBM	3.932	2.959	23.73%	0.450
CatBoost	3.930	2.962	23.81%	0.451
FT-Transformer	4.100	3.079	24.72%	0.403
Tab-MLP	4.088	3.101	25.11%	0.406
3-Model Avg (Best GBM)	3.912	2.942	23.61%	0.456



Key Insights:

- GBM consistently outperforms neural networks on tabular data
- All 3 GBM models perform similarly (RMSE ~3.93)
- 3-Model GBM Average: best ensemble (RMSE: 3.912)
- FT-Transformer & Tab-MLP explored; competitive but not superior
- Neural networks confirm GBM remains SOTA for tabular regression

Explainability: SHAP Feature Importance

Top 10 Features by |SHAP| Importance:

1. FICO Midpoint	SHAP =1.485	Higher FICO -> Lower rate
2. Term (months)	SHAP =1.204	60-month loans higher than 36-month
3. DTI Ratio	SHAP =0.922	Higher DTI -> Higher rate
4. Purpose: Credit Card	SHAP =0.667	Refinancing gets preferential rates
5. All Utilization	SHAP =0.520	Higher credit use -> Higher rate
6. Mo Since Recent IL	SHAP =0.491	Recent accounts increase rate
7. Revolving Util Ratio	SHAP =0.413	Credit card utilization impact
8. Loan Amount	SHAP =0.331	Larger loans slightly higher rates
9. Total Balance ex-Mort	SHAP =0.276	Overall debt burden effect
10. Revolving Balance	SHAP =0.274	Outstanding revolving debt

Business Interpretation

Risk-Based Pricing

- FICO is dominant driver - consistent with industry practice
- Higher risk borrowers correctly assigned higher rates

Behavioral Signals

- Credit utilization captures borrower financial stress
- DTI measures repayment capacity beyond credit score
- Recent inquiries indicate credit-seeking (higher risk)

Loan Structure Effects

- 60-month loans carry ~2-3% higher rates (duration risk)
- Debt consolidation gets preferential rates

Model Reliability

- 5 models tested; GBM ensemble most accurate and stable
- Temporal validation confirms generalization to future loans

Conclusion & Lessons Learned

Summary

- 5 models trained: 3 Gradient Boosting + 2 Neural Networks
- Best: 3-Model GBM Average (RMSE 3.912, MAE 2.942)
- Zero data leakage via temporal split + feature audit

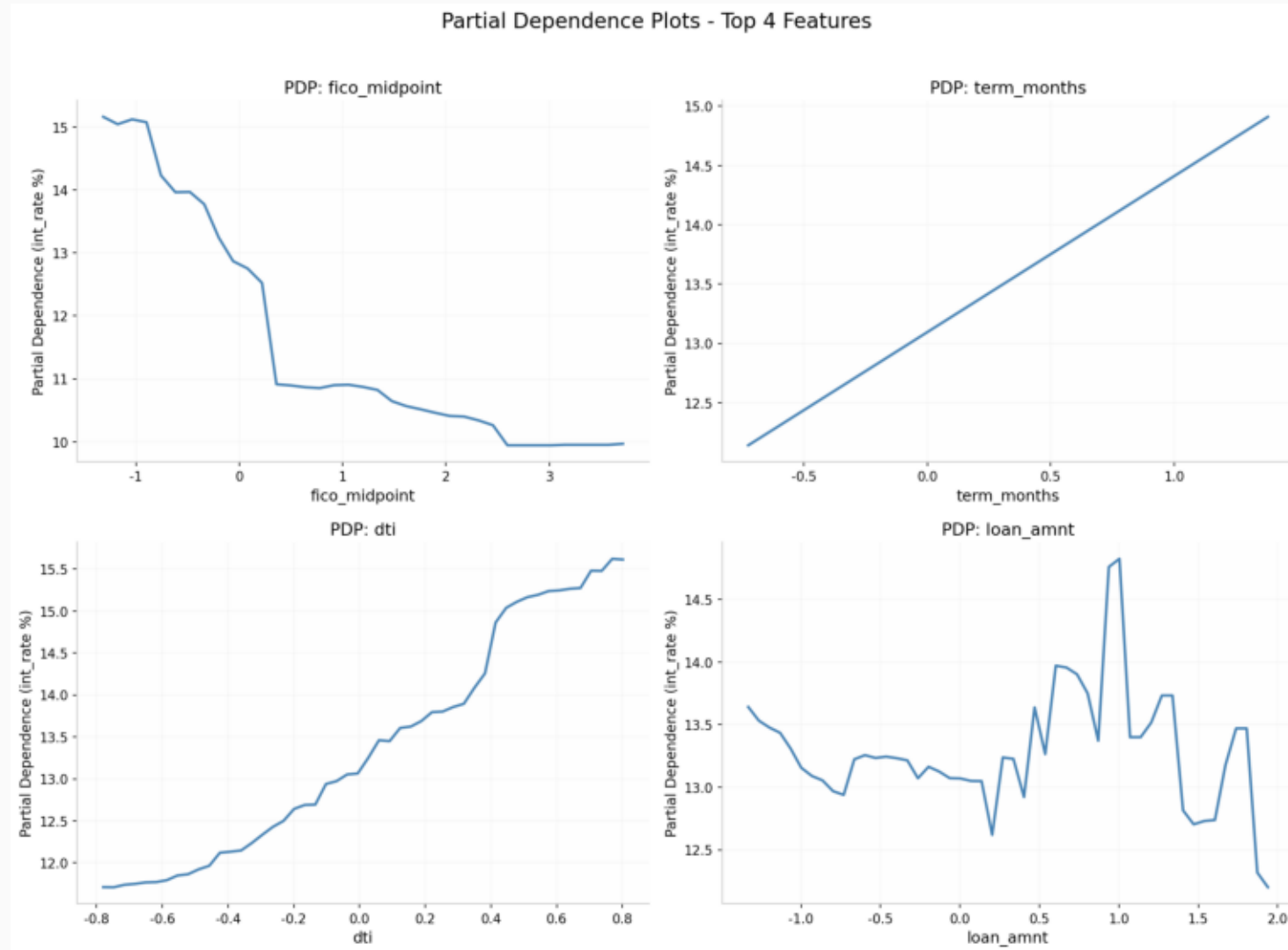
Key Insights

- FICO dominates pricing; DTI and utilization add value
- Simple average ensemble outperforms complex stacking
- NNs explored: FT-Transformer & Tab-MLP competitive but not superior

Lessons Learned

- Data leakage prevention is critical for model validity
- Temporal splits essential for financial time-series
- SHAP explainability builds trust and validates business logic
- For tabular data, gradient boosting remains state-of-the-art

Appendix A: Partial Dependence Plots



Appendix B: Model Diagnostics

